

Total No. of Questions : 8]

P9296

SEAT No. :

[Total No. of Pages : 2

[6181]-522

B.E.(Computers)

EMBEDDED & RTOS

(2015 Pattern) (Semester - II) (410252C)(Elective - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 Q.7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is the use of ROM image in Embedded system? Draw a structure of ROM image and explain various components embedded inside it. [7]
- b) Explain Performance metrics for process of embedded system? [7]
- c) What are types of serial communication in embedded systems? [6]

OR

- Q2)** a) Explain block diagram of Embedded systems? [7]
- b) Explain timing diagram of Analog to Digital Converter (ADC)? [7]
- c) Explain ZigBee protocol in detail? [6]

- Q3)** a) Explain Hard versus soft real-times systems and their timing constraints? [9]
- b) What are Functional parameters and Resource parameters of real-time process? [8]

OR

- Q4)** a) Explain Real-time task and task states with example? [9]
- b) What are Approaches to real-time scheduling? [8]

- Q5)** a) How to Enforcing mutual exclusion and critical sections in embedded systems? [9]
- b) What are Effects of resource contention and resource access control? [8]

OR

P.T.O.

Q6) a) Explain Inter-process communication in RTOS? **[9]**

b) What are Interrupt routines in RTOS environment? **[8]**

Q7) a) Explain synchronization in Real-time Operating system? **[8]**

b) What are different Medium access-control protocols for broadcast networks? **[8]**

OR

Q8) a) Explain Software development process for embedded system? **[8]**

b) What are embedded software development tools. **[8]**

❧ ❧ ❧